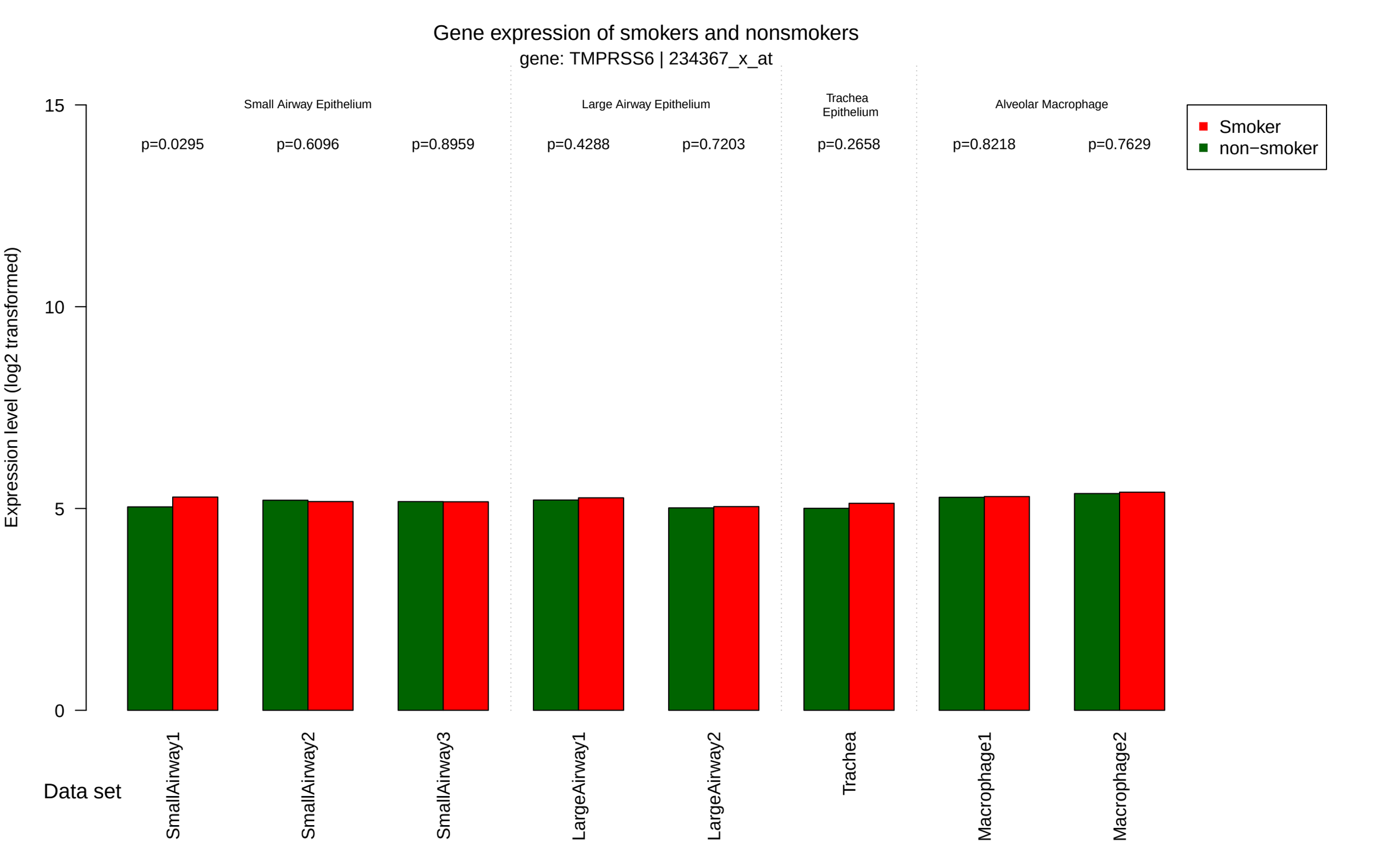
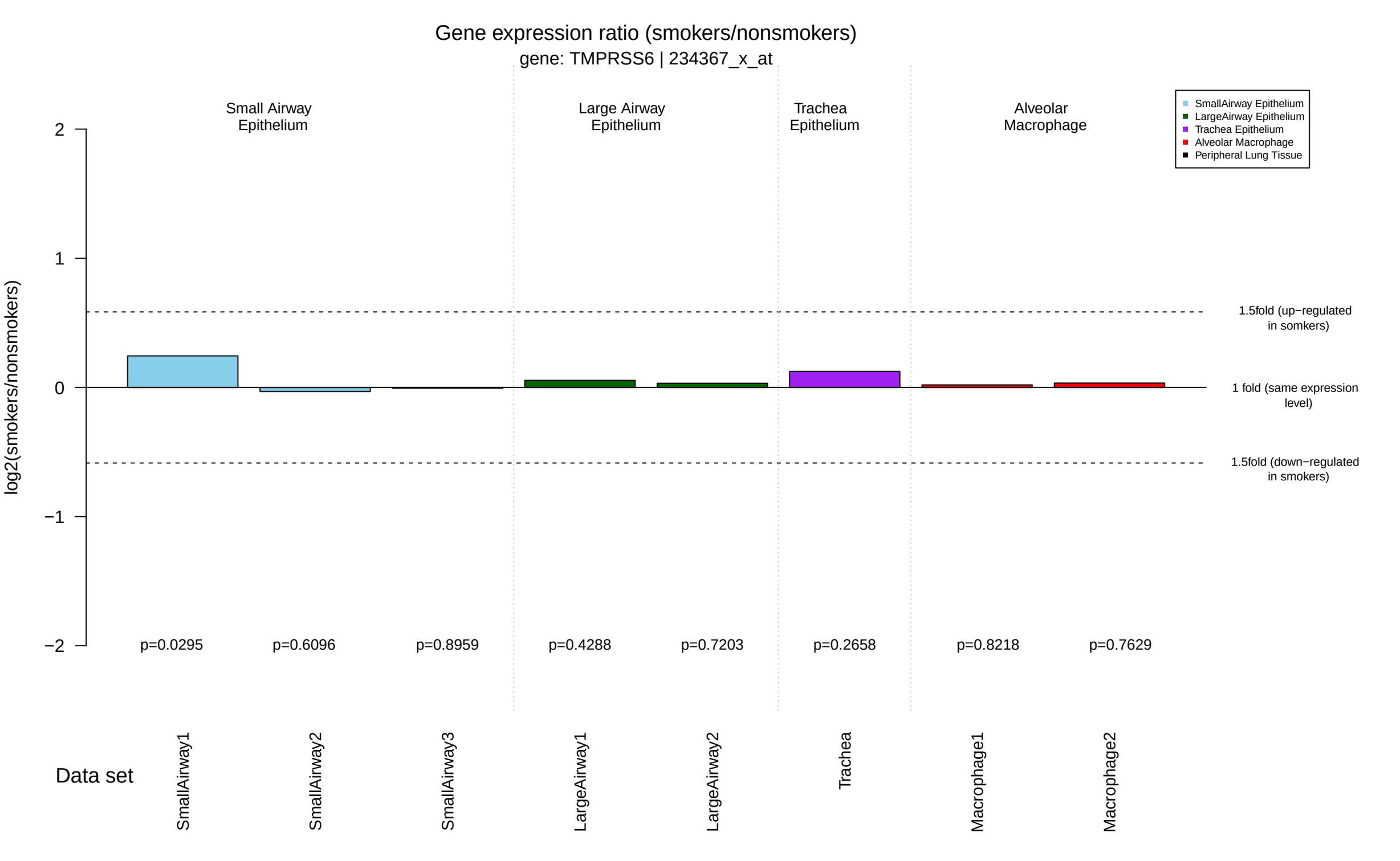
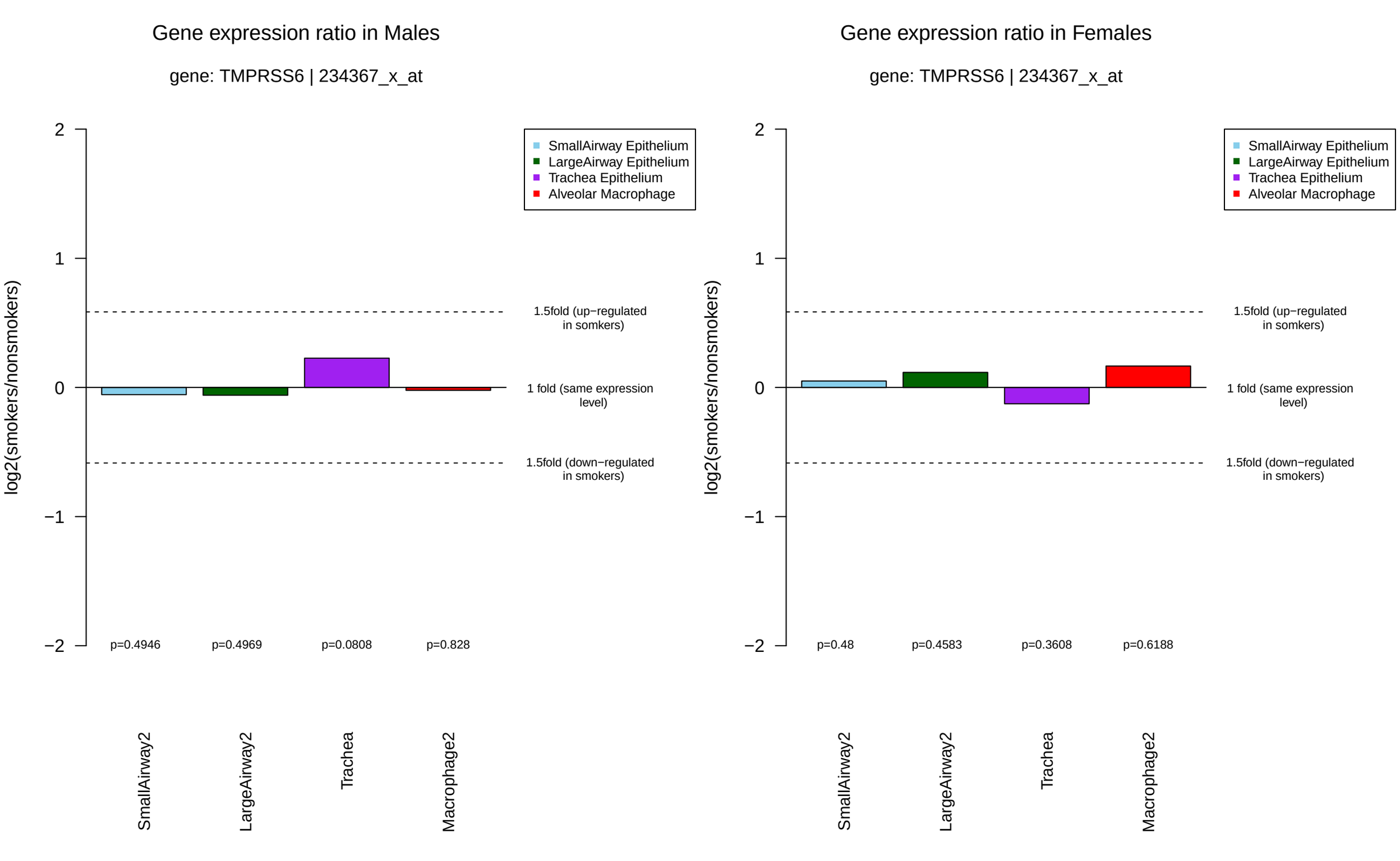
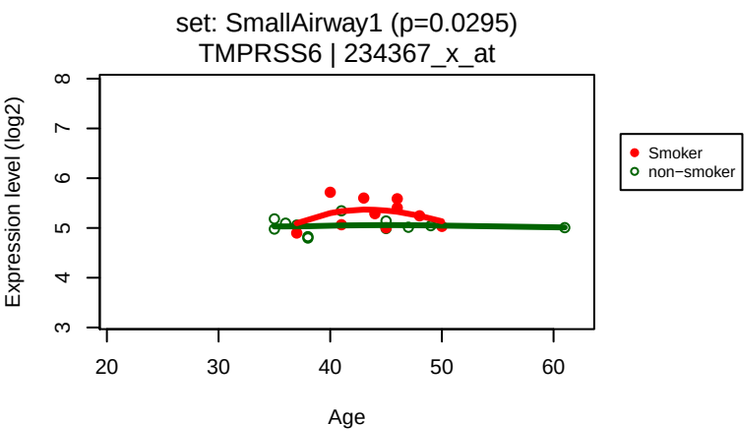
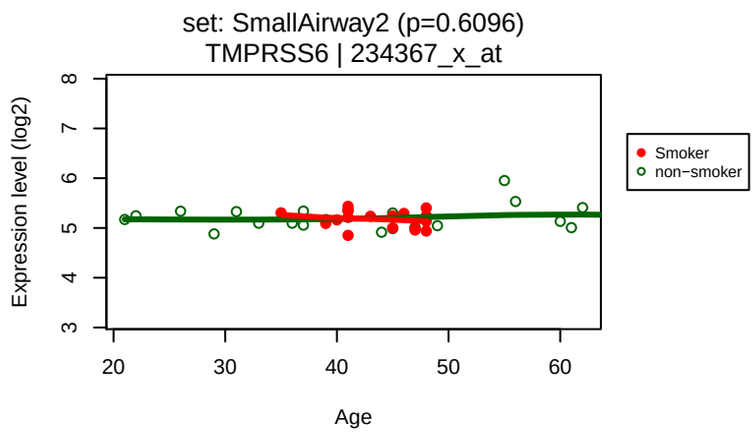
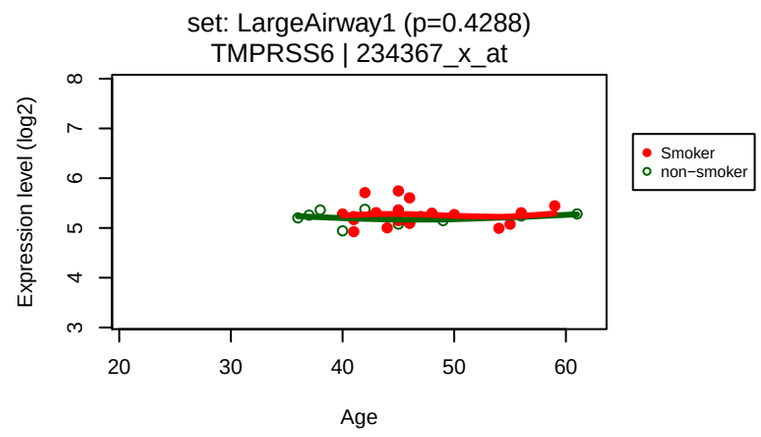

Gene expression of smokers and nonsmokers
gene: TMPRSS6 | 234367_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0295
p=0.6096
p=0.8959
p=0.4288
p=0.7203
p=0.2658
p=0.8218
p=0.7629
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

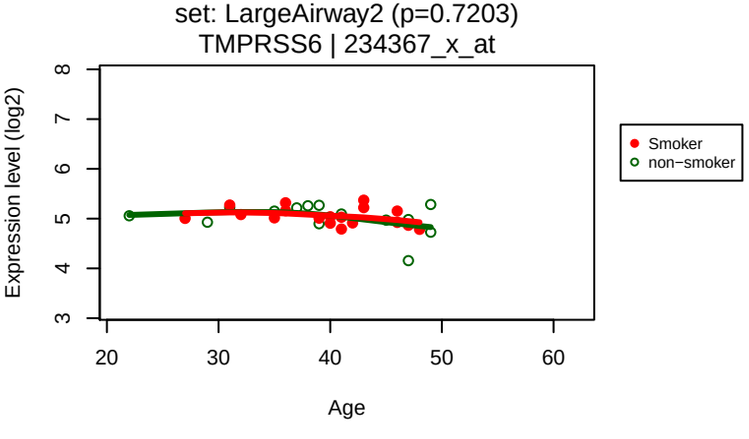

Gene expression ratio (smokers/nonsmokers)
gene: TMPRSS6 | 234367_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0295
p=0.6096
p=0.8959
p=0.4288
p=0.7203
p=0.2658
p=0.8218
p=0.7629
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

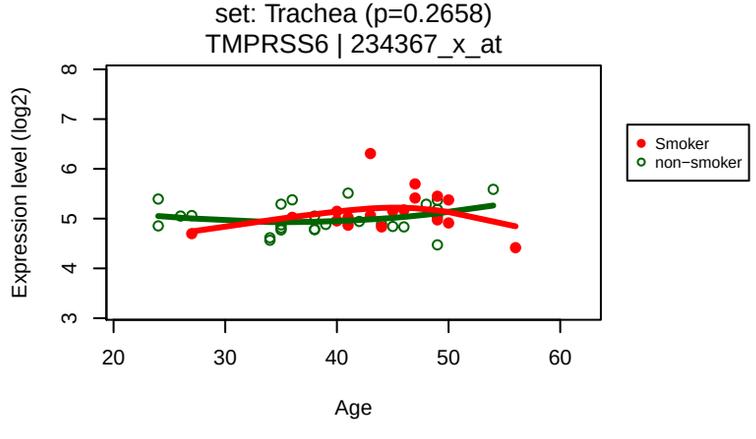

Gene expression ratio in Males
gene: TMPRSS6 | 234367_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4946
p=0.4969
p=0.0808
p=0.828
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMPRSS6 | 234367_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.48
p=0.4583
p=0.3608
p=0.6188
SmallAirway2
LargeAirway2
Trachea
Macrophage2

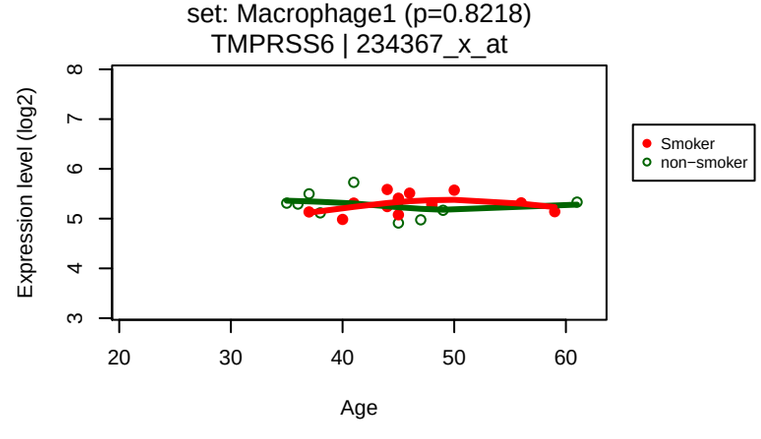

set: SmallAirway1 (p=0.0295)
TMPRSS6 | 234367_x_at
Expression level (log2)
Age
Smoker
non−smoker

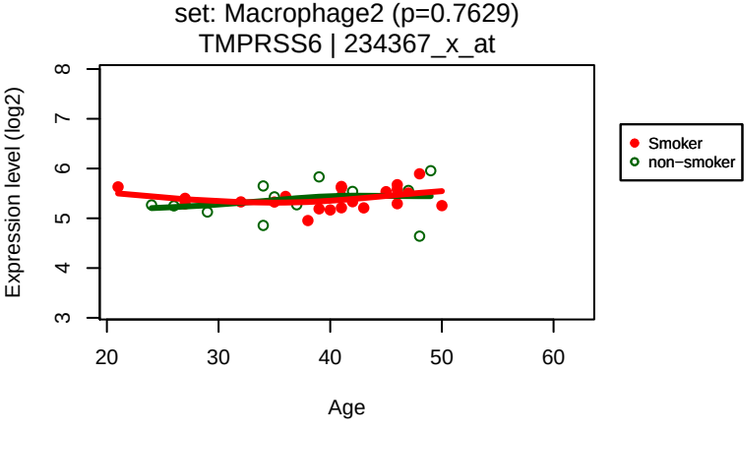

set: SmallAirway2 (p=0.6096)
TMPRSS6 | 234367_x_at
Expression level (log2)
Age
Smoker
non−smoker

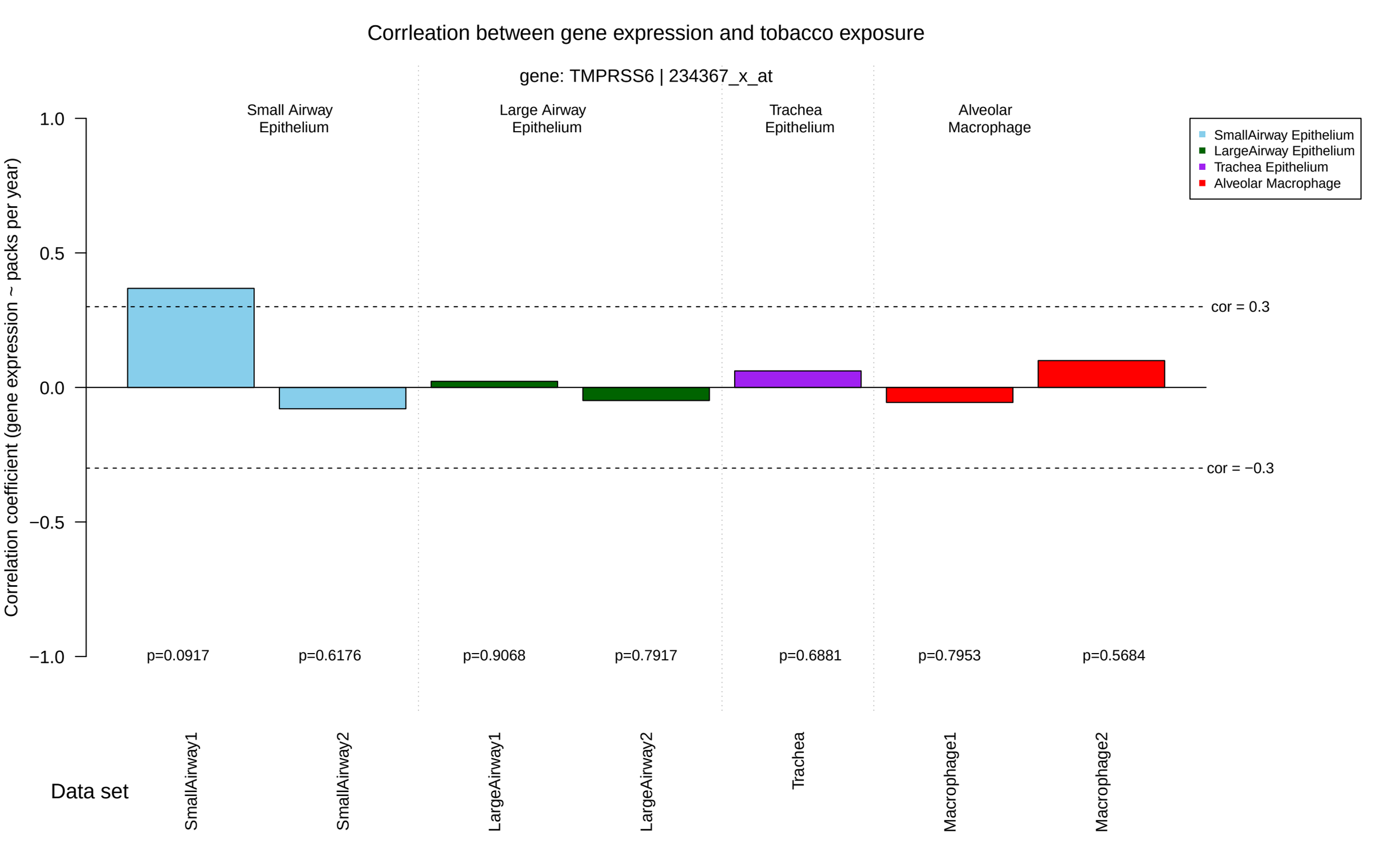

set: LargeAirway1 (p=0.4288)
TMPRSS6 | 234367_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7203)
TMPRSS6 | 234367_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2658)
TMPRSS6 | 234367_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8218)
TMPRSS6 | 234367_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7629)
TMPRSS6 | 234367_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMPRSS6 | 234367_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0917
p=0.6176
p=0.9068
p=0.7917
p=0.6881
p=0.7953
p=0.5684
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2